SCAM SWARM

Autonomous, Evidence-First Multi-Agent Cyber Investigation Platform

"Rejecting ungrounded AI guesses. Scam Swarm combines deterministic OSINT network telemetry with parallel multi-agent adjudication to generate evidence-backed, auditable threat investigation briefs."

THE EVIDENCE INVESTIGATION FLOW

Raw Ingestion → SSRF-Safe OSINT Probes → Structured Evidence Contract → Specialist Swarm Triad → EVIDENCE-REFERENCED INVESTIGATION BRIEF

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THE PROBLEM STATEMENT

Multi-stage digital fraud evades static rules & ungrounded AI.

How psychological coercion, fabricated authority, and dynamic infrastructure defeat single-layer perimeter defenses.

REAL-WORLD CONTEXT

India · Reported Nov 2025

₹24 lakh allegedly lost in a Chennai digital-arrest scam

Cybercriminals allegedly impersonated law-enforcement officers, using intimidation and financial coercion to pressure an elderly victim into transferring funds.

Modus Operandi: Alleged impersonation, intimidation, and financial coercion.

Wider Context: A separately reported alleged ₹303-crore cyber-fraud network illustrates the scale and complexity of financial cybercrime.

Source: *The Times of India*, 21 November 2025

Local Relevance & Separate Macro Ecosystem

OPERATIONAL LIMITATIONS VS. SCAM SWARM

Why traditional triage fails and how Scam Swarm intervenes:

Keyword / Regex Defenses: Miss psychological extortion; fail when messages lack overt malware links. **BLIND (✕)**

Generic AI Prompts: Models hallucinate safety when data is missing; cannot query live registries. **UNGROUNDED (✕)**

Why Scam Swarm Matters:

Identifies coercive language patterns, correlates available technical indicators (domain age, DNS lookup results, including missing or inconclusive data, payment syntax), and produces an **evidence-referenced investigation brief**—without treating an AI-generated suspicion as proof.

CRITICAL DISTINCTION FOR JUDGES: Real-world incidents establish the severe threat landscape, *not* the effectiveness of Scam Swarm. Efficacy is demonstrated separately via empirical benchmark metrics (Slide 7).

PROPOSED SOLUTION

Scam Swarm: Evidence-First Threat Investigation

Decoupling threat analysis into deterministic network collection, specialist agents, and automated verification.

01

DETERMINISTIC FOUNDATION

Live OSINT Probes

- **Live DNS Querying:** Resolves A, AAAA, and MX records to map hosting infrastructure.
- **RDAP Registration Age:** Calculates domain creation age (flags domains < 30 days old as risk signals).
- **Redirect Path Tracing:** Unrolls shortened links hop-by-hop up to a safe limit.
- **Handling Missing Telemetry:** Unresolved lookups are treated as UNKNOWN rather than assumed safe.

02

SPECIALIST AI TRIAD

Specialist Agent Swarm

- **Linguistic Intent Agent:** Analyzes psychological urgency, authority impersonation, and coercion.
- **Infrastructure Agent:** Evaluates ASN reputation, hosting patterns, and redirect hop anomalies.
- **Brand Impersonation Agent:** Evaluates homoglyphic similarity against monitored banking brands.
- **Context Isolation:** Agents run in parallel with distinct prompts to prevent prompt contamination.

03

CLAIM VERIFICATION

Automated Verifier Layer

- **Claim-to-Fact Audit:** Verifies that claims in the synthesized report link directly to probe logs.
- **Ungrounded Claim Detection:** Identifies and filters assertions not supported by recorded telemetry.
- **Bounded Policy Ceilings:** Preserves deterministic policy boundaries against prompt overrides.
- **Structured JSON Output:** Produces formatted evidence logs with timestamps and confidence tags.

CORE PRINCIPLE: Deterministic network probes supply the evidence; multi-agent AI assists in synthesis and explanation.

Five-Layer Architecture & Security Trust Boundaries

Enforcing trust boundaries between deterministic OSINT probes and LLM reasoning.

ZONE A: DETERMINISTIC ENGINE (NODE.JS)

Telemetry Plane

01 / Ingestion & Defanging:
Extracts domains, UPI VPAs, and URLs. Normalizes confusables and neutralizes active links.

02 / SSRF Pre-Resolution Guard:
Pre-resolves hostnames before socket connection, validating IPv4 and IPv6 against private (RFC 1918), loopback, and link-local (169.254) subnets across redirect hops.

03 / Network Probe Suite:
Queries DNS A/AAAA records, RDAP domain registration age, and traces redirects with loop detection up to 5 hops.

04 / Policy Engine:
Calculates bounded 0–100 risk score based on observed evidence; UNKNOWN is not evaluated as CLEAN.

SECURITY GATEWAY

Structured Evidence Contract

```
// Read-Only Telemetry Schema
{
  "target": "hdfc-auth.top",
  "domain_age_days": 2,
  "redirect_hops": 3,
  "risk_ceiling": 88,
  "policy": "CRITICAL",
  "probes": ["dns", "rdap"],
  "tamper_status": "LOCKED"
}
```

Structured read-only contract ensures AI operates on recorded observations rather than assumptions.

ZONE B: AI SWARM (GEMINI/GROQ)

Synthesis Plane

- **Linguistic Intent Agent:**
Identifies psychological coercion hooks, fake urgency, and threat phrasing.
- **Infrastructure Agent:**
Interprets ASN records, hosting reputation, and redirect trajectories.
- **Brand Impersonation Agent:**
Detects visual and homoglyphic similarities against targeted banking entities.
- **Independent Verifier Agent:**
Audits Judge sentences against Zone A raw logs to detect unsupported assertions.

DESIGN BOUNDARY: AI models receive verified telemetry as read-only inputs. Policy bounds remain anchored to measured network data.

PRODUCT CAPABILITIES

Investigation Capabilities for Users and SOC Analysts

Modular investigation tools covering message analysis, visual payloads, evidence graphs, and response steps.

VECTOR INGESTION

Text & Multi-Hop URL Defanging

Traces HTTP redirect paths up to 5 hops, inspects DNS/RDAP records for risk signals, and flags homoglyphic confusables.

VISUAL ANALYSIS

QR Scanning & Screenshot OCR

Extracts embedded URLs from QR codes and reads text from screenshots of suspicious messages or portals using vision models.

FINANCIAL ANALYSIS

UPI Handle Pattern Validation

Checks UPI VPAs against known NPCI provider handle patterns to flag malformed or unverified formats (heuristic indicator, not proof of fraud).

THREAT TOPOLOGY

Interactive Evidence Graph

Interactive canvas DAG connecting analyzed domains, resolved IPs, authoritative nameservers, redirect hops, and risk scores.

INCIDENT RESPONSE

Citizen Advisory & Response Steps

Provides standard emergency guidance, including 1930 Cyber Helpline escalation, initial bank contact steps, and evidence summaries.

EVALUATION LAB

Evaluation Lab & Preset Benchmark

Pre-loaded with 40 labeled adversarial lures from tests/dataset.json for reproducible inspection and benchmark verification.

APPLICATION SCOPE: Helps users quickly inspect suspicious inputs and produces structured evidence summaries for incident reporting.

SYSTEM ENGINEERING

Implementation Architecture & Phased Roadmap

Modular full-stack implementation balancing lightweight deterministic checks with multi-agent orchestration.

FRONTEND CLIENT

React 18 & Vite

- Tailwind CSS responsive UI
- Lucide forensic icon set
- Canvas Evidence Graph DAG
- Responsive layout for mobile/desktop

BACKEND ENGINE

Node.js & Express

- Native dns.promises async queries
- Custom redirect tracer with hop limits
- ipaddr.js SSRF IP validation
- Deterministic policy scoring engine

AI ORCHESTRATION

Gemini & Groq

- Gemini 1.5 Pro / Flash models
- Groq Llama-3-70b for fast triage
- Structured JSON output validation
- Independent Verifier claim check

OSINT SOURCES

Network Registries

- ICANN RDAP bootstrap registry
- Public authoritative DNS resolvers
- NPCI UPI handle pattern specs
- ASN lookup telemetry

PHASED IMPLEMENTATION STATUS

✓ **Stage 1 & 2 (Implemented & Tested):** Ingestion defanging, socket SSRF IP pre-filtering, deterministic policy scoring, and structured evidence contract.

✓ **Stage 3 (Implemented & Tested):** Multi-agent triad (Linguistic, Infra, Brand), Judge synthesis, and automated claim-to-fact Verifier.

→ **Stage 4 (Future Scope):** Sandboxed browser execution for dynamic JavaScript analysis, expanded evaluation corpora, and direct reporting webhooks.

STATUS: Deterministic network probes, SSRF IP validation, and multi-agent claim verification are implemented and passing test suites.

EMPIRICAL VALIDATION

Controlled Evaluation Across 40 Adversarial Samples

Evaluation on tests/dataset.json comparing Heuristic Rules vs Single LLM vs Full Scam Swarm.

ARCHITECTURE TIER	ACC	PREC	REC	SPEC	EVID. COV
Tier 1: Regex & Heuristics	75.0%	100.0%	50.0%	100.0%	60.0%
Tier 2: Single-Pass LLM	**80.0%**	92.9%	**65.0%**	95.0%	65.0%
Tier 3: Full Scam Swarm	**72.5%**	84.6%	55.0%	90.0%	**100.0%**

HONEST BENCHMARK ANALYSIS

Accuracy vs. Evidence Trade-Offs

- **Highest Raw Accuracy in Tier 2:** On this 40-sample set, Tier 2 (Single-Pass LLM) achieves the highest raw accuracy at 80.0%, compared to 72.5% for Scam Swarm.
- **The Evidence Trade-Off:** Tier 2 classifies borderline lures on linguistic cues alone, but 35% of its output lacks verifiable network proof. Tier 3 requires strict evidence links (100% coverage), resulting in a more conservative recall (55.0%).
- **Evaluation Set Limitation:** 40 labeled samples represent a focused preliminary test suite. Broad performance claims require evaluation on much larger, continuously sampled corpora.

Tier 1: Fast Heuristics (50.0% Recall) · **Tier 2: Highest Benchmark Accuracy (80.0%)** · **Tier 3: Full Evidence Provenance (100.0% Coverage)**

CONCLUSION & FUTURE SCOPE

Evidence Before AI Conclusions

Grounding cybersecurity AI in empirical network observations and verifiable claims.

01

CORE APPROACH

Evidence-First Investigation

Separates deterministic network telemetry collection from generative AI synthesis, ensuring claims in generated reports directly reference recorded evidence.

02

EFFICIENCY & PRACTICALITY

Two-Tiered Triage Architecture

Fast deterministic checks handle unambiguous URLs and known formats locally, reserving multi-agent LLM analysis for cases requiring multi-vector correlation.

03

FUTURE SCOPE

Evaluation & Ecosystem Growth

Expanding the benchmark corpus beyond 40 samples, developing client browser integrations for real-time link defanging, and sandboxing dynamic web payloads.

"Evidence Before AI Conclusions: In cybersecurity, threat assessments must be anchored to verifiable evidence and clear confidence limits."

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